Lie Groups — Notes

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We will follow the book "Lectures on Lie groups" by J. Frank Adams through chapter 5.56. Another book relevant to this course is "Naive Lie Theory" by J. Stillwell.

1.1 Introduction

Suppose that V, W are finite dimension vector spaces over $\mathbb{R}$. Notice that in such spaces all norms are equivalent. a function $f : U \to W$ where $U \subseteq V$ is differentiable if for all $x \in U$ there exists a linear map $df\ |_x = f'(x) : V \to W$ such that,

$$f(x+h) = f(x) + (df\ |_x)(h) + o(|h|)$$

Therefore $f' : U \to \mathrm{Hom}(V, W)$ defined by $x \mapsto f'(x)$. We continue to define in such manner the next differentials recursively, $f'' : U \to \mathrm{Hom}(V, \mathrm{Hom}(V, W))$ etc. A function f is C^∞ (smooth) if the derivatives of any order exists and are continuous.

To study manifolds we have to be able to move between manifolds and Euclidean spaces, the map from a local open set in euclidean space into a manifold is called local parameterization or local coordinates.

Definition 1.1 (Chart)**.** Let X be a topological space. **Chart** on X modeled on V, $\varphi : U \to \Omega$ is a homeomorphism such that both $U \subseteq X$ and $\Omega \subseteq V$ are open sets.

In Adams book chart is defined the other way around ($\Omega_\alpha \to X_\alpha$), meaning that Adams chart is equivalent to our definition to parameterization. Compatible charts $\varphi_\alpha, \varphi_\beta$ are charts such that the homeomorphism $\varphi_\beta \circ \varphi_\alpha^{-1}$ (between open subsets of V) is a diffeomorphism.

Definition 1.2 (Atlas)**.** An atlas on X is a family of compatible charts $\{\varphi_\alpha : X_\alpha \to \Omega_\alpha \mid \alpha \in I\}$ for some indices set I, such that $\bigcup_{\alpha \in I} X_\alpha = X$, namely a family of charts that form a cover of X.

Each atlas can be enriched to contain any chart of an open subset of X, thus making it unique maximal atlas. Two atlases are equivalent if their union is an atlas, equivalently if they share a maximal atlas.

Exercise 1.1. Prove that this is an equivalent relation.

Definition 1.3 (Smooth manifold)**.** A (smooth) manifold is a Hausdorff 2nd countable topological space X equipped with an equivalence class of atlases (or a maximal atlas).

Given manifolds X, Y modeled on vector-spaces V, W, $X \times Y$ is a manifold modeled on $V \times W$. $\{\varphi_\alpha : X_\alpha \to \Omega_\alpha\}$, $\{\psi_\beta : Y_\beta \to \Omega'_\beta\}$ are atlases, then $\{\varphi_\alpha \times \psi_\beta : X_\alpha \times Y_\beta \to \Omega_\alpha \times \Omega_{\beta'}\}$ is an atlas in $X \times Y$. Equivalent atlases on X & on Y yield equivalent atlases on $X \times Y$. There exists a unique manifold structure on $X \times Y$ such that $X \times Y \to X$ & $X \times Y \to Y$ are smooth and for any Z manifold and $Z \to X \times Y$ is smooth iff the compositions are smooth.

Definition 1.4 (smooth map between manifolds)**.** Given X, Y are manifolds modeled on V, W and atlases $\{\varphi_\alpha\}$ and $\{\psi_\beta\}$. We say that $f : X \to Y$ is smooth if $\forall \alpha, \beta,\ \varphi_\beta \circ f \circ \psi_\alpha^{-1}$ is smooth.

Definition 1.5 (Lie group)**.** A Lie group G is a set equipped with:

1. a manifold structure
2. a group structure: a group operation $\mu : G \times G \to G$ denoted as $\mu(a, b) = a \cdot b$ and inversion $i : G \to G$ defined as $i(x) = x^{-1}$.

such that μ and i are smooth.

Definition 1.6 (Homomorphism of Lie groups)**.** Let G, H be Lie groups, a function $f : G \to H$ that is smooth and a group Homomorphism is called Lie group Homomorphism.

Example 1.1. $(\mathbb{R}^n, +)$ is a Lie group.

Example 1.2. Observe $(\mathbb{R}^n/\mathbb{Z}^n, +)$, an n-dimensional torus, is a Lie group with parameterizations obtained by restricting the quotient map to small enough subsets of $\mathbb{R}^n$. Recall that in general $\mathbb{R}^n/\mathbb{Z}^n \cong (\mathbb{R}/\mathbb{Z})^n$ it suffices to show that S^1 is a

Lie group. $(S^1, +)$ is indeed a manifold, and it is clear that it is a group as well.

Example 1.3. Let V be a vector-space over $\mathbb{R}$ or $\mathbb{C}$, denoted as Λ, and let $\mathrm{End}(V) = \mathrm{Hom}_\Lambda(V, V)$ the set of linear operators over V. Recall that if $\dim_\Lambda V = n$ then $\mathrm{End}(V) \cong \mathbb{R}^{n\times n}$. Let,

$$\mathrm{Aut}(V) = \{T \in \mathrm{End}(V) \mid \exists S \in \mathrm{End}(V),\ T \circ S = S \circ T = \mathrm{id}\} \subseteq \mathrm{End}(V)$$

be the subset of invertible Λ-linear maps, then $\mathrm{Aut}(V)$ is an open subset of $\mathrm{End}(V)$.

$a, b \mapsto a \cdot b$ as matrix multiplication is a polynomial, and is in particular smooth. It follows from Kramer's rule that $a \mapsto a^{-1}$ the matrix inverse can be obtained by $a^{-1} = \frac{1}{\det a}\,\mathrm{adj}(a)$, and so it's smooth as well.

Remark. For X, Y are manifolds and $f : X \to Y$ the following are equivalent:

1. f is smooth
2. For any $n \in \mathbb{N}$ and open subset $S \subseteq \mathbb{R}^n$ & smooth map $g : S \to X$, the composition $f \circ g$ is smooth
3. For any smooth function $g : Y \to \mathbb{R}$, $f \circ g$ is smooth as well

Exercise 1.2. Prove this

Definition 1.7 (Tangent space)**.** Tangent space to a manifold X at $x \in X$ is defined as,

$$T_xX = \{\gamma : J \to X \mid 0 \in J \subseteq \mathbb{R} \text{ open interval}, \gamma \text{ is smooth}, \gamma(0) = x\}/\sim$$

where we define that $\gamma_1 \sim \gamma_2$ iff for any smooth function $f : U \to \mathbb{R}$ where $x \in U$ is an open neighborhood,

$$\frac{d}{dt}\,|_{t=0}\ f \circ \gamma_1 = \frac{d}{dt}\,|_{t=0}\ f \circ \gamma_2$$

In other words, we identify curves by their "derivative" when passing through x.

Example 1.4. For $x \in X \subseteq V$ open set, namely an open subset of a vector space, $T_xX = V$, as can be seem via $\gamma(t) = x + tv$ for $v \in V$.

Smooth map of manifolds $f : X \to Y$ induces $f' = f_* = df\,|_x \colon T_xX \to T_{f(x)}Y$.

Given a manifold X & atlas $\{\varphi_\alpha : X_\alpha \to \Omega_\alpha\}$,

$$(\varphi_\alpha)^* : T_xX_\alpha \to T_{\varphi_\alpha(x)}\Omega_\alpha = V$$

gives a well defined vector-spaces structure on T_xX.

Definition 1.8 (Lie algebra)**.** Suppose that G is a Lie group and $e \in G$ its unit, we define $L(G) = T_eG$ (as a vector-spaces), Lie algebra of G. Also notated as $\mathrm{Lie}(G)$ or $\mathfrak{g}$.

Let G be a Lie group, $x \in G$, and define left multiplication,

$$L_x : G \to G, \quad g \mapsto xg$$

L_x defines a diffeomorphism $G \to G$ with $e \mapsto x$ hence a linear isomorphism $T_eG \xrightarrow{\sim} T_xG$, then by left trivial injection, $TG \cong G \times L(G)$.

definitions & theorems

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